

ANNEX I

SUMMARY OF PRODUCT CHARACTERISTICS

▼ This medicinal product is subject to additional monitoring. This will allow quick identification of new safety information. Healthcare professionals are asked to report any suspected adverse reactions. See section 4.8 for how to report adverse reactions.

1. NAME OF THE MEDICINAL PRODUCT

Ceplene 0.5 mg/0.5 mL solution for injection

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

One vial of 0.5 mL of solution contains 0.5 mg of histamine dihydrochloride.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection.

Clear, colourless aqueous solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Ceplene maintenance therapy is indicated for adult patients with acute myeloid leukaemia (AML) in first remission concomitantly treated with interleukin-2 (IL-2). The efficacy of Ceplene has not been fully demonstrated in patients older than age 60.

4.2 Posology and method of administration

Ceplene maintenance therapy should be administered following completion of consolidation therapy in patients concomitantly treated with IL-2 under the supervision of a physician experienced in the management of acute myeloid leukaemia.

Posology

For dosing instructions for Ceplene in combination with IL-2, see posology below.

Interleukin-2 (IL-2)

IL-2 is administered twice daily as a subcutaneous injection 1 to 3 minutes prior to the administration of Ceplene; each dose of IL-2 is 16,400 IU/kg (1 µg/kg).

Interleukin-2 (IL-2) is commercially available as a recombinant IL-2; aldesleukin. The dispensing and storage directions in the section 6.6 are specific to aldesleukin.

Ceplene

0.5 mL solution is sufficient for a single dose (see section 6.6).

Ceplene is administered 1 to 3 minutes after each injection of IL-2. Each 0.5 mL Ceplene dose is injected slowly, over 5-15 minutes.

Treatment cycles

Ceplene and IL-2 are administered for 10 treatment cycles: each cycle consists of a treatment period of 21 days (3 weeks) followed by a three-week or six-week treatment-free period.

For cycles 1-3, each cycle consists of 3 weeks of treatment, followed by a 3-week treatment free period. For cycles 4-10, each cycle consists of 3 weeks of treatment, followed by a 6-week treatment free period.

The recommended dosing regimen is presented in Tables 1 and 2.

Table 1: For treatment cycles 1-3 with Ceplene and IL-2

Week number (w)*			Treatment*
Cycle 1	Cycle 2	Cycle 3	
w.1 to w.3 (Days 1-21)	w.7 to w.9 (Days 1-21)	w.13 to w.15 (Days 1-21)	IL-2 16 400 IU/kg followed by 0.5 mL Ceplene. Twice daily.
w.4 to w.6	w.10 to w.12	w.16 to w.18	Treatment-free (3 weeks)

*see dose modification for provisions for the modification to dose and dosage schedule

Table 2: For treatment cycles 4-10 with Ceplene and IL-2, same as for Table 1 above, with the exception of number of cycles and duration of rest periods

Week number (w)*							Treatment*
Cycles							
4	5	6	7	8	9	10	
w.19 to w.21	w.28 to w.30	w.37 to w.39	w.46 to w.48	w.55- to w.57	w.64 to w.66	w.73 to w.75	IL-2 16 400 IU/kg followed by 0.5 mL Ceplene. Twice daily
w.22 to w.27	w.31 to w.36	w.40 to w.45	w.49 to w.54	w.58 to w.63	w.67 to w.72	w.76 to w.81	Treatment-free (6 weeks)

*see dose modification for provisions for the modification to dose and dosage schedule

Dose modification

Patients should be monitored for the expected symptomatic adverse reactions and laboratory changes associated with this treatment. Doses of Ceplene and IL-2 should be modified as necessary based on individual patient tolerance to treatment. It is recommended that dose modifications be addressed early in treatment. The dose reductions can be temporary or permanent. Should Ceplene related toxicities occur (such as hypotension, headache), the injection time can be increased from 5 minutes to a maximum duration of 15 minutes.

For patients experiencing grade 1 toxicity events

No altered dose recommendations with the exception of grade 1 neurologic toxicity and grade 1 generalised toxic dermatitis. For the dose recommendations for these grade 1 toxicity events refer to the relevant sections below:

For patients experiencing grade 1-4 neurologic toxicity

- for grade 1 to 3 toxicity, treatment should be discontinued until grade 0 toxicity event has been achieved. Treatment should then be resumed at a 20% dose reduction for both Ceplene and IL-2.
- for grade 4 toxicity, discontinuation of treatment should be considered.

For patients experiencing grade 1-4 generalised toxic dermatitis

- for grade 1 toxicity, the treatment should be delayed for 48 hours or until all symptoms have been resolved. Treatment should then be resumed using the full dose of Ceplene, but reducing the IL-2 dose by 20%.
- for grade 2 toxicity, the IL-2 dose should be reduced 50% and only increased to full dose if the symptoms do not reappear. Ceplene and IL-2 doses should be separated by 60 minutes, which should be maintained throughout treatment.
- for grade 3 and 4 toxicity, treatment should be discontinued and not resumed until events have been resolved. Treatment should only be resumed after consideration of risk – benefit to the patient.

For patients experiencing grade 2 (including cardiac function, renal, hepatic) toxicity

- treatment should be discontinued until the event has returned to grade 1
- the time of injection of the dose of Ceplene should be extended to a maximum of 15 minutes.
- for cardiac, hepatic or renal toxicities the dose should be reduced by 20% for both Ceplene and IL-2.

For patients experiencing grade 3 and 4 (including hypotension, arrhythmia) toxicities

- treatment should be discontinued until the event is resolved. A maximum delay of one treatment cycle can be considered for the resolution of grade 3 and 4 events.

For persistent hypotension, headache, arrhythmia, cardiac, hepatic and renal toxicities

- the time of injection of the dose of Ceplene should be extended to a maximum of 15 minutes.
- the dose amount of both Ceplene and IL-2 should be reduced by 20%.

Fever

- IL-2 can be discontinued for 24 hours and then restarted at a 20% dose reduction level.

Abnormal WBC counts

- the dose of IL-2 can be reduced by 20% for the remaining duration of the treatment course and if abnormal WBC counts re-occur during the following cycle a permanent IL-2 reduction is recommended.

Localised toxic dermatitis

- treatment should be discontinued until symptoms resolved. Treatment can be resumed by administering Ceplene at the full dose and IL-2 at 50%.

Special populations

Elderly patients

The efficacy of Ceplene has not been fully demonstrated in patients older than age 60.

Renal impairment

Patients with renal impairment may be more sensitive to the blood pressure lowering effects of Ceplene. Although the degree of renal impairment has no demonstrable effect on the pharmacokinetic disposition of Ceplene, caution is warranted when Ceplene is administered to patients with severe renal impairment. However, no Ceplene dose reduction is normally required in renally impaired patients.

Hepatic impairment

Ceplene should be used with caution in patients with moderate to severe hepatic impairment (see section 5.2). Plasma Ceplene levels are higher in patients with moderate and severe liver impairment, and these patient groups tend to experience more tachycardia and lower blood pressure after Ceplene dosing than do patients with normal or mildly affected liver function. Plasma drug levels were not predictive of adverse effects, however, and effects did not correlate closely with drug exposure. Dose reduction of Ceplene is normally not required in hepatically impaired patients, but caution should be used in these patients.

Paediatric population

The safety and efficacy of Ceplene in children below 18 years of age have not yet been established. No data are available.

Method of administration

Ceplene is for subcutaneous use only.

One to 3 minutes after the subcutaneous administration of IL-2 has been completed, Ceplene should be administered by slow subcutaneous injection at a rate not to exceed 0.1 mL (0.1 mg histamine dihydrochloride) per minute. The usual time for administering a 0.5 mL Ceplene dose is 5 minutes. To reduce potential adverse reactions, the administration time may be lengthened to a maximum of 15 minutes, see below. Ceplene can be administered via an ambulatory infusion syringe pump or by controlled manual subcutaneous injection by syringe with a timer.

The first dose of Ceplene and IL-2 on day 1 of the initiation of the first cycle of treatment should be administered in the clinic under direct supervision by a physician. Patient monitoring on day 1 should include vital signs, including pulse, blood pressure and respiratory rate. If the patient experiences a significant change in vital signs, the physician should evaluate the status of the patient and continue to monitor vital signs; these patients should be monitored during subsequent treatments.

Subsequent injections of Ceplene may be self-administered at home by a patient who demonstrates a good understanding of necessary precautions and who has demonstrated adequate injection skills. Injections should be preferably administered in a supervised setting in the presence of an adult family member, friend, or other care provider who is capable of responding appropriately should signs or symptoms of hypotension occur.

The preferred injection areas are the thighs and the abdomen. Ceplene should not be injected into the same anatomic region as IL-2.

The twice daily dosing of IL-2 and Ceplene should be separated by a minimum of 6 hours. Patients should remain at rest for 20 minutes after injection of Ceplene.

For instructions on reconstitution and dilution of Interleukin-2 (aldesleukin) before administration, see commercially available IL-2 SmPC.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.
- Patients with significantly compromised cardiac function, e.g., NYHA Class III/IV.
- Patients receiving systemic steroid therapy, clonidine and H₂ blocking agents.
- Patients who have received an allogenic stem cell transplant.
- During pregnancy.
- During breast feeding.

4.4 Special warnings and special precautions for use

Ceplene should be administered 1 to 3 minutes after IL-2 administration, and not concomitantly. Rapid subcutaneous injection or injection into a vascular space may result in severe hypotension, tachycardia, or syncope.

Treatment with Ceplene in conjunction with IL-2 should be used with caution in patients with poorly compensated cardiac function. Patients with cardiac disease should be evaluated for ventricular ejection fraction and wall function by echocardiography or nuclear medicine stress test and then treated with caution.

Patients should be monitored during treatment for possible clinical complications due to hypotension or hypovolaemia. Ceplene should be administered in the clinic under supervision of the physician on day 1 of the initial treatment cycle. Patient monitoring on day 1 should include vital signs, including pulse, blood pressure and respiratory rate.

Patient monitoring during subsequent treatment days or cycles should be performed as long as the patient continues to experience significant changes in vital signs during administration of Ceplene. If significant hypotension or related symptoms are observed in subsequent treatment cycles, dose reduction should be initiated and if required, administered in hospital until responses to treatment allow for home administration.

Caution should be used for patients with any of the following: symptomatic peripheral arterial disease, past or present peptic or oesophageal ulcer disease with a history of bleeding, clinically significant renal disease and stroke within the last 12 months. Where appropriate, consideration should be made to providing concomitant treatment with a proton pump inhibitor.

Patients with clinically significant infection requiring the use of antibiotics, antifungals, or antivirals, or who have completed prior anti-infectious therapy within 14 days of starting treatment should be treated with caution unless the use of antibiotics and antivirals were for prophylaxis purposes.

Patients with a prior history of autoimmune disease (including systemic lupus, inflammatory bowel disease, psoriasis and rheumatoid arthritis) should be treated with caution.

Monitoring of laboratory test results is recommended including standard haematological and blood chemistry tests.

Patients receiving the following medicinal products should be treated with caution (see section 4.5):

- Beta-blockers or other anti-hypertensive agents.
- H₁ blocking agents and neuroleptics (anti-psychotics) with H₁ receptor blocking properties.
- Tricyclic anti-depressants that may have H₁ and H₂ receptor blocking properties.
- Monoamine oxidase inhibitors and anti-malarial and anti-trypanosomal agents.
- Neuromuscular blocking agents, narcotic analgesics, and various contrast media.

4.5 Interaction with other medicinal products and other forms of interaction

While posology differs, when Ceplene is used in conjunction with IL-2, physicians should also refer to the Summary of Product Characteristics (SmPC) for IL-2 and observe the respective medicinal product interactions.

H₂ receptor antagonists with imidazole structures similar to histamine, e.g., cimetidine, systemic steroids and clonidine, must not be used during treatment with Ceplene (see section 4.3).

Beta-blockers and other anti-hypertensive agents should be used with caution during treatment with Ceplene. Concurrent administration of medicinal products with cardiotoxicity or blood pressure lowering effects may increase the toxicity of Ceplene.

ANNEX II

- A. MANUFACTURER RESPONSIBLE FOR BATCH RELEASE**
- B. CONDITIONS OR RESTRICTIONS REGARDING SUPPLY AND USE**
- C. OTHER CONDITIONS AND REQUIREMENTS OF THE MARKETING AUTHORISATION**
- D. CONDITIONS OR RESTRICTIONS WITH REGARD TO THE SAFE AND EFFECTIVE USE OF THE MEDICINAL PRODUCT**
- E. SPECIFIC OBLIGATION TO COMPLETE POST-AUTHORISATION MEASURES FOR THE MARKETING AUTHORISATION UNDER EXCEPTIONAL CIRCUMSTANCES**